

NAKHON PHANOM AGROMET, Thailand

WMO: 483580

Lat: 17.412N

Lon: 104.779E

Elev: 502

StdP: 14.43

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	52.6	55.9	45.1	45.1	61.4	47.8	50.1	61.3	17.8	75.8	14.3	74.0	0.2	N/A	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.5	98.8	78.2	96.2	78.3	94.2	78.3	82.6	91.1	81.8	89.9	81.0	88.8	3.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.5	161.2	87.0	79.7	157.2	86.0	79.1	153.9	85.1	47.0	91.5	46.1	90.3	45.3	89.2	93.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.4	8.9	7.2	45.9	101.3	4.0	1.9	43.0	102.6	40.7	103.7	38.4	104.8	35.5	106.2		
			45.1	86.1	4.0	3.8	42.1	88.8	39.8	91.1	37.5	93.2	34.6	96.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.6	71.6	75.5	81.1	84.4	84.0	83.3	81.7	81.8	81.9	80.3	77.3	72.2
	DBStd	5.84	5.11	5.89	5.36	4.16	3.12	2.73	2.92	2.43	2.32	2.70	4.00	5.26
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	22	10	3	1	0	0	0	0	0	0	0	0	8
	CDD50	10806	669	715	965	1032	1054	999	982	986	957	939	819	689
	CDD65	5353	213	298	501	582	589	549	517	521	507	474	370	232
	CDH74	54081	1742	2807	5323	6787	6625	6081	5078	5207	5026	4312	3189	1904
Wind	WSAvg	1.5	2.3	2.3	2.0	1.5	1.1	1.0	0.9	0.9	0.8	1.2	1.7	2.2
	PrecAvg	95.4	0.1	0.5	2.0	3.5	10.2	22.2	17.0	26.0	11.7	2.4	0.2	0.1
	PrecMax	111.7	0.4	1.9	4.6	8.0	17.4	40.7	30.4	48.5	23.1	11.5	1.8	1.1
	PrecMin	71.7	0.0	0.0	0.2	0.4	6.5	11.3	10.7	10.8	3.8	0.3	0.0	0.0
	PrecStd	13.7	0.1	0.6	1.5	2.2	2.7	8.9	4.9	10.4	5.2	2.7	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.8	96.5	100.6	102.9	99.2	96.1	94.5	93.2	93.6	93.4	93.1	91.7
		MCWB	72.8	75.0	76.5	78.3	79.9	81.9	80.8	81.4	80.2	78.1	77.3	74.1
	2%	DB	88.1	93.9	97.7	99.9	96.3	93.5	91.4	90.7	91.5	91.4	90.9	89.3
		MCWB	71.6	74.3	76.8	78.6	80.6	80.8	80.7	80.5	79.8	77.0	76.4	73.1
	5%	DB	85.9	91.3	94.9	96.5	93.7	91.3	89.2	89.1	89.9	89.7	89.0	86.6
		MCWB	70.7	73.5	76.4	78.3	80.1	80.6	80.1	80.2	79.7	76.7	75.2	72.1
	10%	DB	83.1	88.1	92.1	93.8	91.2	89.4	87.1	87.3	88.1	88.1	87.0	83.7
		MCWB	69.7	72.4	75.5	78.0	79.6	80.0	79.8	79.7	79.3	76.3	74.2	70.4
	0.4%	WB	77.1	78.2	80.0	82.7	83.4	83.6	82.9	82.7	82.5	80.8	79.4	76.5
		MCDB	84.5	91.6	93.5	94.4	93.9	92.4	90.6	89.7	89.9	89.8	88.7	87.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	74.2	76.8	78.4	81.0	82.0	82.2	81.5	81.6	81.4	79.2	77.9	74.6
		MCDB	83.4	89.6	92.3	93.5	92.4	89.9	88.3	88.3	88.6	87.4	87.7	86.2
	5%	WB	72.5	75.4	77.5	80.0	81.0	81.4	80.7	80.8	80.6	78.2	76.8	73.1
		MCDB	82.3	87.9	90.8	91.9	90.4	88.7	87.1	87.2	87.7	85.9	85.9	84.3
	10%	WB	70.8	73.8	76.6	79.0	80.2	80.7	80.0	80.1	79.9	77.4	75.6	71.4
		MCDB	80.9	84.5	89.0	89.8	88.7	87.5	85.7	85.8	86.2	84.4	83.9	81.7
Mean Daily Temperature Range	MDBR		22.4	22.9	20.2	18.5	14.4	11.5	9.8	10.0	11.7	15.7	19.5	22.5
	5% DB	MCDBR	24.1	25.6	23.0	21.9	17.9	14.4	13.3	13.1	14.8	17.7	20.6	23.4
		MCWBR	9.7	9.0	6.9	6.6	6.0	4.7	4.8	4.9	5.3	6.0	8.0	9.4
	5% WB	MCDBR	21.6	23.2	21.1	19.1	15.7	13.4	12.2	12.3	13.6	15.1	18.7	22.0
		MCWBR	10.1	9.5	7.5	7.0	6.0	5.4	5.2	5.3	5.6	5.7	8.4	9.6
Clear-Sky Solar Irradiance	taub		0.452	0.558	0.662	0.632	0.512	0.474	0.462	0.474	0.504	0.512	0.454	0.455
	taud		2.174	1.895	1.695	1.801	2.154	2.282	2.313	2.266	2.171	2.116	2.251	2.186
	Ebn at Noon		263	240	220	228	254	261	265	263	256	249	260	257
	Edh at Noon		45	62	78	71	49	43	42	44	48	49	41	43
All-Sky Solar Radiation	RadAvg		1482	1586	1581	1690	1613	1477	1365	1364	1433	1499	1476	1397
	RadStd		88	115	66	91	127	141	167	118	128	130	68	55

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	+0.54	N/A	N/A	N/A	+0.36	+0.27	N/A	N/A	N/A	N/A

Nomenclature: See separate page